

Supplementary Material for Topology-Correcting Differentiable Triangle-Soup Reconstruction via Persistent Homology

Anonymous 3DV submission

Paper ID *****

A. The allocation-channel study, in brief

This appendix reports, with their numbers, the three findings of our allocation-channel study — controlled resampling-prior experiments on the same pipeline, scenes, and budgets, conducted before moving topology into the objective — that this paper builds on.

(1) Geometric metrics are topology-blind. Three constructed cases each pit a topologically correct candidate A against a wrong one B whose geometry is bisection-matched, making Chamfer equal by construction (40k samples, metrics normalized by the ground-truth bounding diagonal; Table S1). The discriminating-dimension bottleneck separates every pair — by 35–40× in the H0/H2 cases — while in the first two cases Hausdorff₉₅ actually prefers the topologically wrong candidate. Chamfer cannot tell a benign bump from a destroyed void; the persistence metric can.

(2) The prior arms, and why voids are “width-primary”. The study biases DiffSoup’s own resampler with a pre-computed importance field built from the target’s significant features. Each significant feature f (lifetime p_f above the significance threshold) contributes Gaussian kernels at the coordinates C_f of its birth/death-simplex vertices — the loop-filling triangle, void-filling tetrahedron, or component-merging edge — sized by the feature’s own death scale:

$$\phi_d(q) = \sum_{f: \dim f=d} \sum_{c \in C_f} w_f e^{-\|q-c\|^2/(2\sigma_f^2)}, \quad w_f = p_f/s^2, \\ \sigma_f = s \operatorname{clip}(\sqrt{\operatorname{death}_f}, 3r_{\text{med}}, \tfrac{1}{2}\operatorname{diag}), \quad (1)$$

evaluated on a dense target-surface sample, normalized per dimension to $[0, 1]$ (99.5th percentile), combined by max; a query takes its nearest surface sample’s value.

The spread knob s widens each kernel while preserving its injected surface mass ($\sigma \rightarrow s\sigma$, $w \rightarrow w/s^2$, so $w\sigma^2$ is s -invariant): $s=1$ is the concentrated prior (B2), $s=3$ the spread prior (B4) — the field C5 reuses. The field enters resampling at strict budget parity through two levers: prune protection — faces drop in order of $\text{vis} + \lambda Q_{0.9}(\text{vis}) \cdot \text{imp}$, removing exactly the required count — and respawn bias — top-importance visible faces subdivide first, then the protective order re-prunes to the cap. Remaining arms: B0 baseline; B1 = B2 applied at initialization only; B3/B5 random-center non-topological controls width-matched to B2/B4 respectively. On the void class (Table S2; $N=1200$, five paired seeds) B4 cuts the bottleneck 33–53% below baseline — but the width-matched random control B5 recovers most of that gain, leaving only a small, shape-dependent topological residual (cylinder 4.6σ , sphere 2.2σ , cube tie); one-shot placement washes out entirely ($B1 \approx B0$, .0546 vs .0535 on the sphere). Hence the verdict quoted in the main paper’s introduction: the prior’s topological value is carried largely by its spatial width.

(3) No prior shape repairs loops. On the torus ($N=700$, five paired seeds, true $\beta_1=2$) every arm fails the topology-specificity test: the best condition is the non-topological wide control B5 (.0267), ahead of spread-topological B4 (.0316), baseline (.0409), and narrow random B3 (.0397); the concentrated topological field B2 is worst (.0425) and manufactures phantom handles — final significant-H1 count 4.4 against the true 2.0, with the worst Chamfer (1.32 vs baseline 1.01). Spreading merely stops the harm; it never beats the random control. That failure is the opening premise of the main paper, and the class the loss channel wins there (main paper §5.1, Table 1).

Table S1. Blindness probes: A topologically correct, B wrong, geometry bisection-matched so Chamfer is equal by construction. Bottleneck to ground truth in the discriminating dimension.

case (disc. dim)	Cham.% A = B	bott. A	bott. B	ratio
thin handle, genus 0→1 (H1)	0.916	≈ 0	.0302	—
bridge merges two components (H0)	0.628	.0014	.0574	40×
punctured enclosed void (H2)	0.977	.0040	.1385	35×

Table S2. Allocation study, void class (H2), $N=1200$, five paired seeds: tail bottleneck to target (lower is better). Spread beats concentrated; the width-matched random control (B5) recovers most of the spread gain.

shape	B0	B2 conc.	B3 narrow rand.	B4 spread	B5 wide rand.
sphere	.0535	.0438	.0440	.0357	.0374
cube	.0579	.0409	.0403	.0273	.0270
cylinder	.0545	.0440	.0477	.0360	.0402

Table S3. Significant bars (H1 / H2) on the clean target surfaces vs. sampling density M ; floor = $6r_{\text{med}}(M)$, seed 0.

	$M=512$	$M=1024$	$M=2048$	$M=4096$
sphere	0 / 1	0 / 1	0 / 1	0 / 1
torus	1 / 0	1 / 0	2 / 0	2 / 1
double torus	0 / 0	2 / 0	2 / 0	4 / 0

B. Diagnostics: density sensitivity and diagram trajectories

Sampling density vs. measurability. Table S3 counts the significant bars of the clean target surfaces across sampling densities under the self-calibrating floor $6r_{\text{med}}(M)$ of the main paper’s §6. It confirms the three density claims made in the paper: the sphere’s void clears the floor at every tested density; the torus needs $M=2048$ before its second loop is measurable, and its own void becomes measurable only at $M=4096$ (hence the H1 restriction at $M=2048$); and the double torus’s tube loops cross the floor between $M=2048$ and $M=4096$ — consistent with the $M \gtrsim 3 \times 10^3$ estimate there — so all four loops are measurable at $M=4096$.

Diagram trajectories: the pinning mechanism, visualized. Fig. S1 shows the live H2 diagrams of one sphere seed under C0/C1/C2 at four training steps. All three arms start with the void essentially on target (initialization samples the SfM point cloud), and photometric training alone drags it away (the baseline settles $\approx .06$ below). The loss arm dips identically while the ramp is inactive, then pulls the void back to within $\approx .016$ of

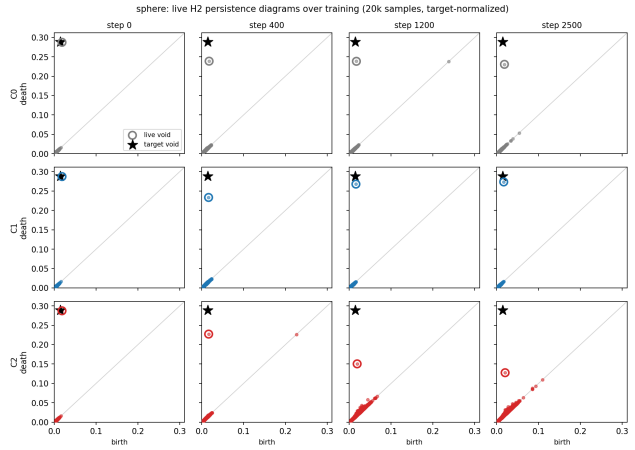


Figure S1. Live H2 persistence diagrams over training (one sphere seed; 20k samples, target-normalized; star = target void bar, circle = most persistent live bar). Top, baseline: the void drifts off target. Middle, loss: the void returns to the target once the ramp activates. Bottom, repulsion control: the void is pinned and dragged away while near-diagonal phantom bars accumulate.

target — matching its measured tail. The control arm is qualitatively different: the void’s death is dragged progressively down ($\approx .23 \rightarrow .15 \rightarrow .13$) while a carpet of phantom mid-persistence bars accumulates near the diagonal — the repulsion-equilibrium signature discussed in the main paper’s §6.

The genus-2 stretch case, in full. On the double torus every arm — baseline, loss, control — returns a bottleneck of exactly .0264 (sd 0 over seeds), and Wasserstein is equally pinned (.0527–.0528): the two tube loops are unexpressable at feasible budgets (replicating the allocation study’s floor, §A) and dominate both diagram metrics regardless of guidance. The loss adds no phantoms (#sig $\beta_1 = 2$ every run) at marginally best Chamfer.

Sensor-noise internals (C7/C7h). Recruitment absorbs the injected noise: at $\sigma=1\%$ the torus’s second loop is sub-threshold at every refresh, with zero unreached targets. Calibration is the noisiest stage (λ_{peak}

Table S4. Tom-yum pot challenge inventory: what makes the shape hard, and the measured consequence in this study.

challenge	measured consequence
raw artist export: 9 open shells, mixed quad/12-gon faces, 50 boundary edges after exact weld, 232 non-manifold junction edges	no trustworthy ground-truth topology as exported; ingest-and-certify required (weld, offset-solidify, drop 804 enclosed pocket shells)
certified shell: one body, genus 3, $\beta=(1,6,1)$ exact, 190,990 vertices	ground truth known by certificate (ex-body, act simplicial homology; watertightness cross-check; bit-identical rebuilds), not by construction
six thin handle/vent loops	every H1 loop sits below the significance floor at every working density — H1 is unusable as the loss observable
narrow flue mouth, capping at small α	the flue interior reads as an enclosed on-axis chamber — an H2 void, first measurable at $M=8192$ (margin $1.20-1.23\times$, five sampling seeds); the floor rule picks class and density prospectively (Table S5)
in-loop outcome	C1 cuts the void's error $3.9\times$ at Chamfer parity; the repulsion control destroys it ($\beta_2 1\rightarrow 0$, all seeds, $1.36\times$ worse Chamfer) — main paper Table 2

Table S5. Tom-yum pot staircase (seed 0; the $M=8192$ margin spans seeds 0–4): significant-bar signature $(\beta_0, \beta_1, \beta_2)$ of the certified shell vs. sampling density M , floor = $6r_{\text{med}}(M)$. The bundle is built at the first density where the discriminating feature clears ($M=8192$: exactly one significant bar, H2, stable to re-sampling at $\sim 10^{-5}$).

	$M=2048$	$M=4096$	$M=8192$	$M=20000$
sig. bars	(1, 0, 0)	(1, 0, 0)	(1, 0, 1)	(1, 0, 1)

spreads $\approx 3\times$ across seeds), yet tails stay tight — consistent with the flat ρ response reported in the main paper.

The tom-yum pot: challenge inventory and prospective floor protocol. The generality wave's artist mesh (main paper §4, §5.2) compounds the failure modes real scans will bring. Table S4 inventories them with the measured consequence of each — every row is backed by the ingest certificate or by a measurement in this study, none is asserted. Table S5 records the staircase behind the flue row: the certified shell's significant-bar signature across sampling density, which fixed the observable class (H2) and the bundle density ($M=8192$) before any training run.

C. Additional result figures and the repair gate 122

Fig. S2 shows the generality-wave training trajectories (main paper §5.2, Table 2); Fig. S3 the per-seed tail spread behind the sphere row of main-paper Table 1; Fig. S4 the torus significant-count check behind §5.1's zero-phantom-handles claim. 123 124 125 126 127

The stage-3a gate (main paper §5). Fig. S5 shows three of the four repair probes run before integration: the loss alone (no photometric term, Adam directly on point positions) must repair defective clouds. The predicted location-blind pathology of recruitment did not materialize, for a mechanism reason worth stating: tearing a 2-manifold shell cannot grow a component bar past the local detour scale (the tug is abandoned), while cutting the 1-D bridge raises the recruited bar's death monotonically toward target — the runaway concentrates on the only profitable direction. 128 129 130 131 132 133 134 135 136 137 138

D. A decade of reconstruction, cut by optimization philosophy 139 140

The main paper's related work argues one sentence that this appendix substantiates in breadth: reconstruction methods of the differentiable decade, whatever their representation, optimize geometric objectives (photometric fidelity plus geometric regularizers) or steer sampling under such an objective — topology, when correct, is correct by construction, by extraction, or by accident, never because the training objective measured it. Table S6 tabulates representative methods under this cut, including the nearest concurrent topology-aware entries; the subsections survey each family and the adjacent literatures this paper draws machinery or discipline from. 141 142 143 144 145 146 147 148 149 150 151 152 153

D.1. Geometry-oriented: implicit fields 154

Occupancy and signed-distance learning deliver closed surfaces by construction [54, 61, 62], and a fitting stack matured around the level set: Eikonal regularization [25], surface rendering [60, 83], SDF-based volume rendering [79, 84], geometric priors [87], high-resolution and accelerated variants [51, 57, 80], meshing for deployment [85], and kernel-based point-cloud fitting [39]. Across all of them the training objective is photometric or geometric; the guarantee of consistent topology comes from the representation, its correctness (the right genus for the scene) is emergent, and the price is dense field evaluation plus post-hoc extraction — the cost regime the soup family exists to avoid. 155 156 157 158 159 160 161 162 163 164 165 166 167

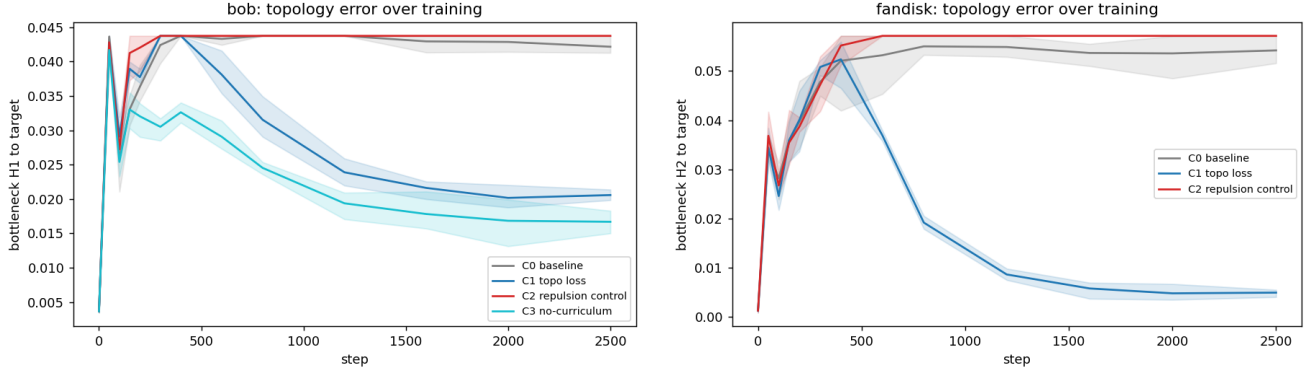


Figure S2. Generality trajectories (line = seed mean, band = seed min-max). Left: bob (genus 1) — the H1 result off the analytic family: the loss halves the loop error with #sig H1 = 2 in every seed, while the control rides the baseline after collapsing a loop. Right: fandisk (CAD) — the largest effect measured in this study (10.4 $\times$).

Table S6. Representative reconstruction methods by optimization philosophy. “Topology in objective?” asks whether the training objective measures a topological quantity of the reconstructed surface (not whether the representation constrains topology, and not image-space summaries). Bottom block: the nearest topology-aware concurrent work (§D.7).

method	repr.	philosophy	topology in objective?	topology from
NeRF [55]	density field	geometry	no	emergent
OccNet [54]	occupancy	geometry	no	construction (closed iso)
DeepSDF [61]	SDF	geometry	no	construction (closed iso)
NeuS [79]	SDF	geometry	no	construction (closed iso)
Neuralangelo [51]	SDF	geometry	no	construction (closed iso)
Pixel2Mesh [78]	template mesh	geometry	no	frozen at template genus
Point2Mesh [30]	template mesh	geometry	no	frozen at template genus
Mesh R-CNN [24]	voxels $\rightarrow$ mesh	geometry	no	fixed by voxelization
DMTet [71]	grid + marching	geometry	no	extraction over the grid
nvdiffr [58]	grid + marching	geometry	no	extraction over the grid
FlexiCubes [72]	grid + marching	geometry	no	extraction over the grid
Shape-as-Points [63]	points $\rightarrow$ implicit	geometry	no	construction (closed iso)
AtlasNet [26]	patch soup	geometry	no	unmeasured
DSS [86]	point soup	geometry	no	unmeasured
3DGS [47]	Gaussian soup	geometry	no	unmeasured
2DGS [37]	surfel soup	geometry	no	unmeasured
SuGaR [27]	Gaussians $\rightarrow$ mesh	geometry	no	post-hoc extraction
MiLo [28]	Gaussians+mesh	geometry	no	in-loop extraction, unmeasured
Triangle Splatting [31]	triangle soup	geometry	no	unmeasured
Radiant Tri. Soup [8]	triangle soup	geometry	no	unmeasured (proximity forces)
Mini-Splatting [19]	Gaussian soup	sampling	no	unmeasured (budgeted resample)
DiffSoup [76]	triangle soup	geometry	no	unmeasured
DiffSoup + prior (§A)	triangle soup	sampling	no (steers only)	unmeasured; fails on loops
Topology-GS [73]	Gaussian soup	topology (image)	rendered images only	surface unmeasured
Gao et al. [22, 23]	watertight mesh	geometry	no (topology a priori)	template genus / camera prior
STITCH [41]	implicit (points)	topology	yes (β_0 , live surface)	construction + constraint
this work	triangle soup	topology	yes (live-surface persistence)	measured, optimized, budgeted

D.2. Geometry-oriented: extraction, grids, and templates

Where the surface is explicit, its topology is decided by an extraction algorithm or frozen by a template. Marching Cubes and Dual Contouring fix per-cell connectivity by convention [43, 52]; Poisson reconstruction extracts an indicator iso-surface, watertight but with

emergent genus [44, 45]; differentiable iso-surfacing makes extraction trainable without making topology a target [21, 58, 65, 71, 72]; learned tessellations move the per-cell decision into a network [14, 15]; and template pipelines fix topology up front and then only deform [24, 30, 78]. Robust meshing gives the same guarantee-by-construction for simulation inputs [36].

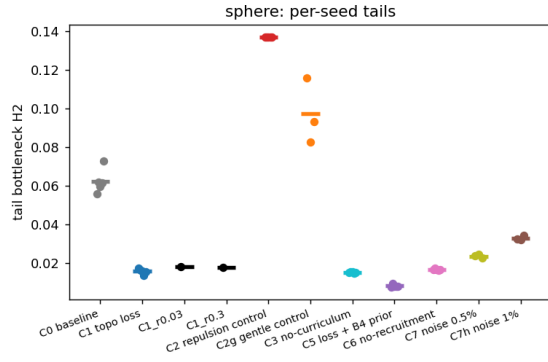


Figure S3. Sphere, per-seed tail bottlenecks (dots) with arm means (bars). Every loss arm sits far below baseline with tight seed spread; the two control strengths bracket baseline from above; the ρ ablation arms are indistinguishable from $\rho=0.1$.

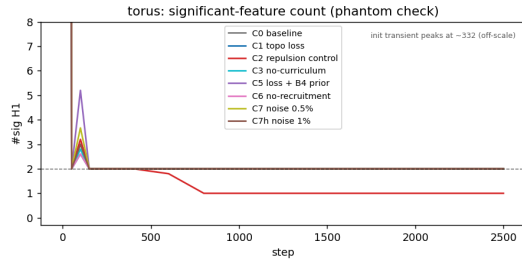


Figure S4. Torus phantom check (main paper §5.1): significant-H1 count over training (seed means; dashed line: true $\beta_1=2$). After the brief resampling transient near step 100, every loss arm sits exactly on two significant loops — zero phantom handles — while the repulsion control (C2) collapses a loop by step ~ 800 and never recovers it.

In every case topology is an input or an artifact of the algorithm — measured nowhere, optimized never.

D.3. Geometry-oriented: point splatting and Gaussian soups

The connectivity-free line runs from EWA surface splatting [92] through neural point rendering [2, 67, 82] to the Gaussian-splatting era [47, 88] — Pulsar’s abstract advertises the family’s freedom from mesh “topology problems” explicitly [49]. Its surface-minded members regularize geometry ever harder — flattened surfels [18, 37], planar compression [13], opacity-field extraction [89], alignment-driven meshing [27] — and MILO now extracts a mesh differentially at every training iteration [28]; triangle soups follow the same pattern [8, 31, 76]. Yet even with connectivity in the loop, no member of the family measures a topological quantity of what it builds: the objectives remain photomet-

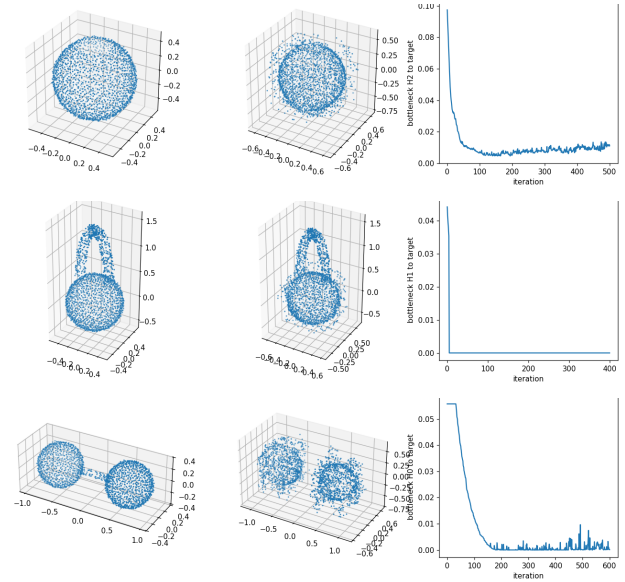


Figure S5. The stage-3a gate: the loss alone (Adam on raw points, no photometric term) repairs defective clouds; each row shows initial cloud, final cloud, and bottleneck-to-target trajectory. Top (T1): punctured sphere — the matched birth term closes the void (8.8 $\times$). Middle (T2): spurious handle — the diagonal term kills the extra H1 bar. Bottom (T4): bridged merge with no significant live H0 bar — recruitment severs the bridge (450 $\times$, β_0 1 $\rightarrow$ 2), the case where plain optimal matching probably has zero gradient. (T3, mis-spaced components, 1229 $\times$: omitted for space.)

ric fidelity plus geometric regularity.

D.4. Sampling-oriented: steering where a fixed objective spends

The second occupied family changes where primitives go, not what the objective asks of them: split-and-prune densification in 3DGS [47], budget-constrained Gaussian resampling [19], importance-driven respawn in DiffSoup [76] — and the topology-inspired allocation priors of our own study (§A), which put the family’s implicit hypothesis to a controlled test and returned the null result the main paper builds on. Steering allocation is a real lever for geometric coverage; the study shows it is structurally the wrong lever for one-dimensional topology.

D.5. Topology as machinery: persistence in and around learning

The ingredients for topology-aware objectives have been maturing for two decades: persistence computation at scale [5, 53, 91], practical diagram metrics [46], stability theory [17], and vectorized representations [1, 7] on the computational side; trainable

diagram layers [9, 32], filtration learning [33], and general differentiability frameworks [10, 50, 59] on the gradient side (surveyed in [11, 12]). In-loop topological losses already win in adjacent domains — representation learning [56], tubular and topologically faithful segmentation [16, 35, 74, 75], microscopy volume prediction [77], point-cloud repair [6, 64] — and differentiable non-persistence signals exist too [66]. What had not happened is the transfer of this machinery into image-based surface reconstruction.

D.6. Topology outside the loop: repair and benchmarks

Before differentiable pipelines, topology was corrected after the fact: spurious-handle removal [29, 81], general mesh repair [3, 42, 70] (surveyed in [4]), and soup-to-manifold conversion [36, 38]; Thingi10K quantifies how pervasive soup-like defects are in the wild [90] — our tom-yum pot’s raw export is a typical specimen. The evaluation culture matches: standard pipelines and benchmarks — COLMAP [68, 69], PMVS [20], DTU [40], Tanks and Temples [48] — score accuracy and completeness by point distances; none carries a topological-correctness metric, the blindness our Table S1 quantifies on controlled probes.

D.7. The empty cell, and its nearest neighbours

Concurrent work has begun to press on the gap from several directions, and the distinctions matter. Topology-GS [73] puts a persistence term inside Gaussian-splatting training — but on barcodes of rendered images, a topological-perceptual loss; the surface’s diagram is never measured, and densification adds primitives rather than budgeting them. The high-genus inverse-rendering line [22, 23] reconstructs meshes whose genus is guaranteed by a Gauss–Bonnet-matched template or guides camera placement with homology computed once on a reference shape — topology as a prior, assumed rather than measured, with ground-truth genus required up front. STITCH [41] does measure a differentiable persistence quantity on a live implicit surface — enforcing a single connected component — but from point-cloud input: no images, no primitives, no budget. Topology also enters generation as conditioning features [34], again outside the reconstruction objective. Partition the three requirements — (i) a topological quantity of the evolving surface measured differentiably in the objective, (ii) image-based reconstruction through a renderer, (iii) a fixed primitive budget — and each neighbour holds at most one; none holds all three. That cell — the bottom row of Table S6 — is what the main paper fills.

References

- [1] Henry Adams, Tegan Emerson, Michael Kirby, Rachel Neville, Chris Peterson, Patrick Shipman, Sofya Chepushtanova, Eric Hanson, Francis Motta, and Lori Ziegelmeier. Persistence images: A stable vector representation of persistent homology. *Journal of Machine Learning Research*, 18(8):1–35, 2017. 5
- [2] Kara-Ali Aliev, Artem Sevastopolsky, Maria Kolos, Dmitry Ulyanov, and Victor Lempitsky. Neural point-based graphics. In *European Conference on Computer Vision (ECCV)*, pages 696–712, 2020. 5
- [3] Marco Attene. A lightweight approach to repairing digitized polygon meshes. *The Visual Computer*, 26(11):1393–1406, 2010. 6
- [4] Marco Attene, Marcel Campen, and Leif Kobbelt. Polygon mesh repairing: An application perspective. *ACM Computing Surveys*, 45(2), 2013. 6
- [5] Ulrich Bauer. Ripser: Efficient computation of Vietoris–Rips persistence barcodes. *Journal of Applied and Computational Topology*, 5:391–423, 2021. 5
- [6] Rickard Brüel-Gabrielsson, Vignesh Ganapathi-Subramanian, Primoz Skraba, and Leonidas J. Guibas. Topology-aware surface reconstruction for point clouds. *Computer Graphics Forum*, 39(5):197–207, 2020. 6
- [7] Peter Bubenik. Statistical topological data analysis using persistence landscapes. *Journal of Machine Learning Research*, 16(3):77–102, 2015. 5
- [8] Nathaniel Burgdorfer and Philippos Mordohai. Radiant triangle soup with soft connectivity forces for 3D reconstruction and novel view synthesis. *arXiv:2505.23642*, 2025. 4, 5
- [9] Mathieu Carrière, Frédéric Chazal, Yuichi Ike, Théo Lacombe, Martin Royer, and Yuhei Umeda. PersLay: A neural network layer for persistence diagrams and new graph topological signatures. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 2786–2796, 2020. 6
- [10] Mathieu Carrière, Frédéric Chazal, Marc Glisse, Yuichi Ike, Hariprasad Kannan, and Yuhei Umeda. Optimizing persistent homology based functions. In *International Conference on Machine Learning (ICML)*, 2021. 6
- [11] Mathieu Carrière, Yuichi Ike, Théo Lacombe, and Naoki Nishikawa. Persistence-based topological optimization: A survey. *arXiv:2603.24613*, 2026. 6
- [12] Frédéric Chazal and Bertrand Michel. An introduction to topological data analysis: Fundamental and practical aspects for data scientists. *Frontiers in Artificial Intelligence*, 4:667963, 2021. 6
- [13] Danpeng Chen, Hai Li, Weicai Ye, Yifan Wang, Weijian Xie, Shangjin Zhai, Nan Wang, Haomin Liu, Hujun Bao, and Guofeng Zhang. PGSR: Planar-based Gaussian splatting for efficient and high-fidelity surface reconstruction. *IEEE Transactions on Visualization and Computer Graphics*, 2024. 5
- [14] Zhiqin Chen and Hao Zhang. Neural marching cubes. *ACM Transactions on Graphics*, 40(6), 2021. 4
- [15] Zhiqin Chen, Andrea Tagliasacchi, and Hao Zhang. BSP-Net: Generating compact meshes via binary space partitioning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020. 4
- [16] James R. Clough, Nicholas Byrne, Ilkay Oksuz, Veronika A. Zimmer, Julia A. Schnabel, and Andrew P. King. A topological loss function for deep-learning based image segmentation using persistent homology. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(12):8766–8778, 2022. 6
- [17] David Cohen-Steiner, Herbert Edelsbrunner, and John Harer. Stability of persistence diagrams. *Discrete & Computational Geometry*, 37(1):103–120, 2007. 5
- [18] Pinxuan Dai, Jiamin Xu, Wenxiang Xie, Xinguo Liu, Huamin Wang, and Weiwei Xu. High-quality surface reconstruction using Gaussian surfels. In *ACM SIGGRAPH 2024 Conference Papers*, 2024. 5
- [19] Guangchi Fang and Bing Wang. Mini-splatting: Representing scenes with a constrained number of Gaussians. In *European Conference on Computer Vision (ECCV)*, 2024. 4, 5
- [20] Yasutaka Furukawa and Jean Ponce. Accurate, dense, and robust multiview stereopsis. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 32(8):1362–1376, 2010. 6
- [21] Jun Gao, Wenzheng Chen, Tommy Xiang, Clement Fuji Tsang, Alec Jacobson, Morgan McGuire, and Sanja Fidler. Learning deformable tetrahedral meshes for 3D reconstruction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. 4
- [22] Xiang Gao, Xinmu Wang, Xiaolong Wu, Jiazhi Li, Jingyu Shi, Yu Guo, Yuanpeng Liu, Xiyun Song, Heather Yu, Zongfang Lin, and Xianfeng David Gu. Inverse rendering for high-genus surface meshes from multi-view images. *arXiv:2511.18680*, 2025. 3DV 2026. 4, 6
- [23] Xiang Gao, Xinmu Wang, Yuanpeng Liu, Yue Wang, Junqi Huang, Wei Chen, and Xianfeng Gu. Inverse rendering for high-genus 3D surface meshes from multi-view images with persistent homology priors. *arXiv:2601.12155*, 2026. ICASSP 2026. 4, 6
- [24] Georgia Gkioxari, Jitendra Malik, and Justin Johnson. Mesh R-CNN. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9784–9794, 2019. 4
- [25] Amos Gropp, Lior Yariv, Niv Haim, Matan Atzmon, and Yaron Lipman. Implicit geometric regularization for learning shapes. In *International Conference on Machine Learning (ICML)*, 2020. 3
- [26] Thibault Groueix, Matthew Fisher, Vladimir G. Kim, Bryan C. Russell, and Mathieu Aubry. AtlasNet: A papier-mâché approach to learning 3D surface generation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018. 4

- [27] Antoine Guédon and Vincent Lepetit. SuGaR: Surface-aligned Gaussian splatting for efficient 3D mesh reconstruction and high-quality mesh rendering. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 5354–5363, 2024. 4, 5
- [28] Antoine Guédon, Diego Gomez, Nissim Maruani, Bingchen Gong, George Drettakis, and Maks Ovsjanikov. MILO: Mesh-in-the-loop Gaussian splatting for detailed and efficient surface reconstruction. *ACM Transactions on Graphics*, 44(6), 2025. 4, 5
- [29] Igor Guskov and Zoë Wood. Topological noise removal. In *Proceedings of Graphics Interface*, pages 19–26, 2001. 6
- [30] Rana Hanocka, Gal Metzer, Raja Giryes, and Daniel Cohen-Or. Point2Mesh: A self-prior for deformable meshes. *ACM Transactions on Graphics*, 39(4), 2020. 4
- [31] Jan Held, Renaud Vandeghen, Adrien Delière, Abdullah Hamdi, Silvio Giancola, Anthony Cioppa, Andrea Vedaldi, Bernard Ghanem, Andrea Tagliasacchi, and Marc Van Droogenbroeck. Triangle splatting for real-time radiance field rendering. *arXiv:2505.19175*, 2025. 4, 5
- [32] Christoph Hofer, Roland Kwitt, Marc Niethammer, and Andreas Uhl. Deep learning with topological signatures. In *Advances in Neural Information Processing Systems (NIPS)*, 2017. 6
- [33] Christoph Hofer, Florian Graf, Bastian Rieck, Marc Niethammer, and Roland Kwitt. Graph filtration learning. In *International Conference on Machine Learning (ICML)*, pages 4314–4323, 2020. 6
- [34] Jiangbei Hu, Ben Fei, Baixin Xu, Fei Hou, Weidong Yang, Shengfa Wang, Na Lei, Chen Qian, and Ying He. Topology-aware latent diffusion for 3D shape generation. *arXiv:2401.17603*, 2024. 6
- [35] Xiaoling Hu, Fuxin Li, Dimitris Samaras, and Chao Chen. Topology-preserving deep image segmentation. In *Advances in Neural Information Processing Systems 32 (NeurIPS)*, 2019. 6
- [36] Yixin Hu, Qingnan Zhou, Xifeng Gao, Alec Jacobson, Denis Zorin, and Daniele Panozzo. Tetrahedral meshing in the wild. *ACM Transactions on Graphics*, 37(4), 2018. 4, 6
- [37] Binbin Huang, Zehao Yu, Anpei Chen, Andreas Geiger, and Shenghua Gao. 2D Gaussian splatting for geometrically accurate radiance fields. In *ACM SIGGRAPH 2024 Conference Papers*, 2024. 4, 5
- [38] Jingwei Huang, Yichao Zhou, and Leonidas Guibas. ManifoldPlus: A robust and scalable watertight manifold surface generation method for triangle soups. *arXiv:2005.11621*, 2020. 6
- [39] Jiahui Huang, Zan Gojcic, Matan Atzmon, Or Litany, Sanja Fidler, and Francis Williams. Neural kernel surface reconstruction. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 3
- [40] Rasmus Jensen, Anders Dahl, George Vogiatzis, Engin Tola, and Henrik Aanaes. Large scale multi-view stereopsis evaluation. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 406–413, 2014. 6
- [41] Anushrut Jignasu, Ethan Herron, Zhanhong Jiang, Soumik Sarkar, Chinmay Hegde, Baskar Ganapathysubramanian, Aditya Balu, and Adarsh Krishnamurthy. STITCH: Surface reconstruction using implicit neural representations with topology constraints and persistent homology. *arXiv:2412.18696*, 2024. 4, 6
- [42] Tao Ju. Robust repair of polygonal models. *ACM Transactions on Graphics*, 23(3):888–895, 2004. 6
- [43] Tao Ju, Frank Losasso, Scott Schaefer, and Joe Warren. Dual contouring of Hermite data. In *Proceedings of the 29th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH)*, pages 339–346, 2002. 4
- [44] Michael Kazhdan and Hugues Hoppe. Screened Poisson surface reconstruction. *ACM Transactions on Graphics*, 32(3), 2013. 4
- [45] Michael Kazhdan, Matthew Bolitho, and Hugues Hoppe. Poisson surface reconstruction. In *Proceedings of the Fourth Eurographics Symposium on Geometry Processing (SGP)*, pages 61–70, 2006. 4
- [46] Michael Kerber, Dmitriy Morozov, and Arnur Nigmatov. Geometry helps to compare persistence diagrams. *ACM Journal of Experimental Algorithmics*, 22, 2017. 5
- [47] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3D Gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42(4):139:1–139:14, 2023. 4, 5
- [48] Arno Knapitsch, Jaesik Park, Qian-Yi Zhou, and Vladlen Koltun. Tanks and temples: Benchmarking large-scale scene reconstruction. *ACM Transactions on Graphics*, 36(4), 2017. 6
- [49] Christoph Lassner and Michael Zollhöfer. Pulsar: Efficient sphere-based neural rendering. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1440–1449, 2021. 5
- [50] Jacob Leygonie, Steve Oudot, and Ulrike Tillmann. A framework for differential calculus on persistence barcodes. *Foundations of Computational Mathematics*, 22:1069–1131, 2022. 6
- [51] Zhaoshuo Li, Thomas Müller, Alex Evans, Russell H. Taylor, Mathias Unberath, Ming-Yu Liu, and Chen-Hsuan Lin. Neuralangelo: High-fidelity neural surface reconstruction. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 3, 4
- [52] William E. Lorensen and Harvey E. Cline. Marching cubes: A high resolution 3D surface construction algorithm. In *Proceedings of the 14th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH)*, pages 163–169, 1987. 4

- [53] Clément Maria, Jean-Daniel Boissonnat, Marc Glisse, and Mariette Yvinec. The Gudhi library: Simplicial complexes and persistent homology. In *International Congress on Mathematical Software (ICMS)*, pages 167–174, 2014. 5
- [54] Lars Mescheder, Michael Oechsle, Michael Niemeyer, Sebastian Nowozin, and Andreas Geiger. Occupancy networks: Learning 3D reconstruction in function space. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019. 3, 4
- [55] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tan-cik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. NeRF: Representing scenes as neural radiance fields for view synthesis. In *European Conference on Computer Vision (ECCV)*, 2020. 4
- [56] Michael Moor, Max Horn, Bastian Rieck, and Karsten Borgwardt. Topological autoencoders. In *International Conference on Machine Learning (ICML)*, pages 7045–7054, 2020. 6
- [57] Thomas Müller, Alex Evans, Christoph Schied, and Alexander Keller. Instant neural graphics primitives with a multiresolution hash encoding. *ACM Transactions on Graphics*, 41(4), 2022. 3
- [58] Jacob Munkberg, Jon Hasselgren, Tianchang Shen, Jun Gao, Wenzheng Chen, Alex Evans, Thomas Müller, and Sanja Fidler. Extracting triangular 3D models, materials, and lighting from images. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8280–8290, 2022. 4
- [59] Arnur Nigmatov and Dmitriy Morozov. Topological optimization with big steps. *Discrete & Computational Geometry*, 2024. 6
- [60] Michael Oechsle, Songyou Peng, and Andreas Geiger. UNISURF: Unifying neural implicit surfaces and radiance fields for multi-view reconstruction. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021. 3
- [61] Jeong Joon Park, Peter Florence, Julian Straub, Richard Newcombe, and Steven Lovegrove. DeepSDF: Learning continuous signed distance functions for shape representation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 165–174, 2019. 3, 4
- [62] Songyou Peng, Michael Niemeyer, Lars Mescheder, Marc Pollefeys, and Andreas Geiger. Convolutional occupancy networks. In *European Conference on Computer Vision (ECCV)*, 2020. 3
- [63] Songyou Peng, Chiyu Jiang, Yiyi Liao, Michael Niemeyer, Marc Pollefeys, and Andreas Geiger. Shape as points: A differentiable Poisson solver. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. 4
- [64] Adrien Poulenard, Primož Skraba, and Maks Ovsjanikov. Topological function optimization for continuous shape matching. *Computer Graphics Forum*, 37(5):13–25, 2018. 6
- [65] Edoardo Remelli, Artem Lukoianov, Stephan R. Richter, Benoît Guillard, Timur Bagautdinov, Pierre Baque, and Pascal Fua. MeshSDF: Differentiable iso-surface extraction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. 4
- [66] Ernst Roell and Bastian Rieck. Differentiable euler characteristic transforms for shape classification. In *International Conference on Learning Representations (ICLR)*, 2024. 6
- [67] Darius Rückert, Linus Franke, and Marc Stamminger. ADOP: Approximate differentiable one-pixel point rendering. *ACM Transactions on Graphics*, 41(4), 2022. 5
- [68] Johannes L. Schönberger and Jan-Michael Frahm. Structure-from-motion revisited. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4104–4113, 2016. 6
- [69] Johannes L. Schönberger, Enliang Zheng, Jan-Michael Frahm, and Marc Pollefeys. Pixelwise view selection for unstructured multi-view stereo. In *European Conference on Computer Vision (ECCV)*, pages 501–518, 2016. 6
- [70] Andrei Sharf, Thomas Lewiner, Gil Shklarski, Sivan Toledo, and Daniel Cohen-Or. Interactive topology-aware surface reconstruction. *ACM Transactions on Graphics*, 26(3):43:1–43:10, 2007. 6
- [71] Tianchang Shen, Jun Gao, Kangxue Yin, Ming-Yu Liu, and Sanja Fidler. Deep marching tetrahedra: a hybrid representation for high-resolution 3D shape synthesis. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. 4
- [72] Tianchang Shen, Jacob Munkberg, Jon Hasselgren, Kangxue Yin, Zian Wang, Wenzheng Chen, Zan Gojcic, Sanja Fidler, Nicholas Sharp, and Jun Gao. Flexible isosurface extraction for gradient-based mesh optimization. *ACM Transactions on Graphics*, 42(4), 2023. 4
- [73] Tianqi Shen, Shaohua Liu, Jiaqi Feng, Ziyi Ma, and Ning An. Topology-aware 3D Gaussian splatting: Leveraging persistent homology for optimized structural integrity. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 6823–6832, 2025. 4, 6
- [74] Suprosanna Shit, Johannes C. Paetzold, Anjany Sekuboyina, Ivan Ezhov, Alexander Unger, Andrey Zhylka, Josien P. W. Pluim, Ulrich Bauer, and Bjoern H. Menze. cDice – a novel topology-preserving loss function for tubular structure segmentation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021. 6
- [75] Nico Stucki, Johannes C. Paetzold, Suprosanna Shit, Bjoern Menze, and Ulrich Bauer. Topologically faithful image segmentation via induced matching of persistence barcodes. In *International Conference on Machine Learning (ICML)*, pages 32698–32727, 2023. 6
- [76] Kenji Tojo, Bernd Bickel, and Nobuyuki Umetani. DiffSoup: Direct differentiable rasterization of triangle soup for extreme radiance field simplification. In *IEEE/CVF Conference on Computer Vision and Pat-*

- tern Recognition (CVPR), 2026. arXiv:2603.27151. 4, 5
- [77] Dominik J. E. Waibel, Scott Atwell, Matthias Meier, Carsten Marr, and Bastian Rieck. Capturing shape information with multi-scale topological loss terms for 3D reconstruction. In Medical Image Computing and Computer Assisted Intervention (MICCAI), pages 150–159, 2022. 6
- [78] Nanyang Wang, Yinda Zhang, Zhuwen Li, Yanwei Fu, Wei Liu, and Yu-Gang Jiang. Pixel2Mesh: Generating 3D mesh models from single RGB images. In European Conference on Computer Vision (ECCV), 2018. 4
- [79] Peng Wang, Lingjie Liu, Yuan Liu, Christian Theobalt, Taku Komura, and Wenping Wang. NeuS: Learning neural implicit surfaces by volume rendering for multi-view reconstruction. In Advances in Neural Information Processing Systems (NeurIPS), 2021. 3, 4
- [80] Yiming Wang, Qin Han, Marc Habermann, Kostas Daniilidis, Christian Theobalt, and Lingjie Liu. NeuS2: Fast learning of neural implicit surfaces for multi-view reconstruction. In IEEE/CVF International Conference on Computer Vision (ICCV), 2023. 3
- [81] Zoë Wood, Hugues Hoppe, Mathieu Desbrun, and Peter Schröder. Removing excess topology from isosurfaces. ACM Transactions on Graphics, 23(2):190–208, 2004. 6
- [82] Qiangeng Xu, Zexiang Xu, Julien Philip, Sai Bi, Zhixin Shu, Kalyan Sunkavalli, and Ulrich Neumann. Point-NeRF: Point-based neural radiance fields. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pages 5438–5448, 2022. 5
- [83] Lior Yariv, Yoni Kasten, Dror Moran, Meirav Galun, Matan Atzmon, Ronen Basri, and Yaron Lipman. Multiview neural surface reconstruction by disentangling geometry and appearance. In Advances in Neural Information Processing Systems (NeurIPS), 2020. 3
- [84] Lior Yariv, Jiatao Gu, Yoni Kasten, and Yaron Lipman. Volume rendering of neural implicit surfaces. In Advances in Neural Information Processing Systems (NeurIPS), 2021. 3
- [85] Lior Yariv, Peter Hedman, Christian Reiser, Dor Verbin, Pratul P. Srinivasan, Richard Szeliski, Jonathan T. Barron, and Ben Mildenhall. BakedSDF: Meshing neural SDFs for real-time view synthesis. In ACM SIGGRAPH 2023 Conference Proceedings, 2023. 3
- [86] Wang Yifan, Felice Serena, Shihao Wu, Cengiz Öztireli, and Olga Sorkine-Hornung. Differentiable surface splatting for point-based geometry processing. ACM Transactions on Graphics, 38(6), 2019. 4
- [87] Zehao Yu, Songyou Peng, Michael Niemeyer, Torsten Sattler, and Andreas Geiger. MonoSDF: Exploring monocular geometric cues for neural implicit surface reconstruction. In Advances in Neural Information Processing Systems (NeurIPS), 2022. 3
- [88] Zehao Yu, Anpei Chen, Binbin Huang, Torsten Sattler, and Andreas Geiger. Mip-splatting: Alias-free 3D Gaussian splatting. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024. 5
- [89] Zehao Yu, Torsten Sattler, and Andreas Geiger. Gaussian opacity fields: Efficient adaptive surface reconstruction in unbounded scenes. ACM Transactions on Graphics, 43(6), 2024. 5
- [90] Qingnan Zhou and Alec Jacobson. Thingi10K: A dataset of 10,000 3D-printing models. arXiv:1605.04797, 2016. 6
- [91] Afra Zomorodian and Gunnar Carlsson. Computing persistent homology. Discrete & Computational Geometry, 33:249–274, 2005. 5
- [92] Matthias Zwicker, Hanspeter Pfister, Jeroen van Baar, and Markus Gross. Surface splatting. In Proceedings of the 28th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH), pages 371–378, 2001. 5